

# Chapter 105 — Evaluation and Management of Women With Urinary Incontinence and Pelvic Organ Prolapse

## DEFINITION AND IMPACT OF URINARY INCONTINENCE AND PELVIC ORGAN PROLAPSE

### *Pelvic floor disorders.*

- **Female pelvic floor disorders (PFDs)** comprise urinary incontinence (UI), faecal incontinence (FI), pelvic organ prolapse (POP), sensory and emptying abnormalities of the lower urinary tract, and defecatory dysfunction.
- **About one woman in four has POP, UI or FI** — UI the commonest (about one in six), then FI, then symptomatic POP.
- **Prevalence is predicted to rise** because both drivers — **ageing and obesity** — are increasing in the population.

### *Urinary incontinence.*

- **International Continence Society (ICS) definition: the complaint of involuntary loss of urine.** Principal types: stress (SUI), urgency (UUI) and mixed (MUI); others are listed in Table 105.1.
- **Reported prevalence varies widely** — about one-third of women over 60 worldwide on meta-analysis; up to four-fifths of **frail elderly** women; US National Health and Nutrition Examination Survey (NHANES) cycles show a rising trend, to over half. **Variation reflects** lack of age standardisation, culture, willingness to report and inconsistent definitions.
- **Risk factors for all types: age, obesity, smoking, comorbidity and postmenopausal hormone therapy.** Vaginal delivery → more SUI. Lower UUI with higher household income, physical activity and caesarean (not vaginal) delivery.
- **Pregnancy:** UI rises from about one in ten in the first trimester to about one in three in the third.
- **Race:** Asian American women have the lowest UI prevalence; White and Mexican-American women have **more SUI** than Black women; Black women have **more UUI and detrusor overactivity (DO)**. Disparities in access to care probably distort these estimates.
- **Subtype order (current US data): isolated SUI > UUI > MUI.** Earlier data ranked MUI second. SUI and MUI prevalence stable 2005–2018; **UUI rising in women aged 60 and over.**
- **Burden:** direct costs in billions of dollars annually, rising as more women are treated; **cumulative risk of SUI surgery 13.6% by age 80.**
- **Functional impact:** physical, social, psychological and sexual. Many women stop exercising because of UI → risk of **diabetes and cardiovascular disease**; UI impairs intimacy and leads to avoidance of intercourse.

### *Pelvic organ prolapse.*

- **ICS definition: descent of one or more of the anterior vaginal wall, posterior vaginal wall, uterus and cervix, or vaginal vault** (cuff after hysterectomy).
- **Commonest site: anterior wall > posterior wall > apex.**
- **Anatomical vs clinical POP:** about **half of parous women** have some degree of POP on examination, but the **diagnosis requires symptoms** attributable to it.
- **Symptoms:** vaginal bulge, pelvic pressure, difficulty voiding or defecating, and **splinting** (hand or fingers in the vagina) for comfort or evacuation.
- **Vaginal bulge is the symptom most specific to POP and the one most likely to resolve after repair.**

- **Symptomatic POP** (asked as "Do you see or feel a bulge?") affects about 3% of US women, stable 2005–2010; highest in White women, lower in Hispanic, Black and Asian American women.
- **Cumulative risk of POP surgery 12.6% by age 80**; direct surgical costs run to millions of dollars annually.

#### RISK FACTORS FOR PELVIC ORGAN PROLAPSE

| Category                   | Factors                                                                                                                               |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Primary POP only           | Increasing age; higher body mass index; higher birth weight; levator muscle defects; lower education (less than high school); poverty |
| Recurrence only            | <b>Younger age</b> ; higher preoperative grade (POP Quantification System [POP-Q] stage 3–4)                                          |
| Both primary and recurrent | Vaginal delivery; parity; levator hiatal hernia (ballooning)                                                                          |
| Protective (primary POP)   | <b>Caesarean delivery; smoking</b>                                                                                                    |

- **Trap:** age cuts in opposite directions — **older age predisposes to primary POP, younger age to recurrence** (longer life-time exposure after repair).
- **Trap:** smoking is **protective** for primary POP, although it is a risk factor for UI.

#### Knowledge and care-seeking.

- **Knowledge of UI and POP is poor** among community-dwelling women, more so among non-White women → many **suffer in silence**.
- **Only 18–50% of women with PFDs seek care.** Barriers are multifactorial; misbeliefs call for **culturally competent** education.
- Costs, lost productivity, reduced quality of life (QoL) and inactivity-related illness → need for **trained providers**.

TABLE 105.1 STANDARD INTERNATIONAL CONTINENCE SOCIETY TERMINOLOGY OF URINARY INCONTINENCE SYMPTOMS

| Term                     | Complaint of involuntary loss of urine...                                                 |
|--------------------------|-------------------------------------------------------------------------------------------|
| Urinary incontinence     | ...of any kind (the umbrella definition)                                                  |
| Stress UI                | ...on effort or physical exertion (including sport), sneezing or coughing                 |
| Urgency UI               | ...associated with urgency                                                                |
| Mixed UI                 | ...associated both with urgency and with effort, exertion, sneezing or coughing           |
| Nocturnal enuresis       | Involuntary <i>voiding</i> during the main sleep period (bedwetting)                      |
| Coital UI                | ...during or after vaginal intercourse                                                    |
| Continuous UI            | ...continuously                                                                           |
| Disability-associated UI | ...through functional inability to reach a toilet in time (physical or mental impairment) |
| Insensible UI            | Aware of leakage but <b>unaware of how or when</b> it occurred                            |
| Postural UI              | ...on change of posture (supine or seated → standing)                                     |
| Occult (latent) SUI      | Stress incontinence seen <b>only after reduction of coexistent POP</b>                    |

- **What the table is for:** every term is a *complaint* (symptom), not a urodynamic diagnosis; the categories are defined by the **trigger** of leakage.

- **Traps:** nocturnal enuresis is *voiding*, not leakage; **insensible** (unaware of the moment) differs from **disability-associated** (aware but cannot reach the toilet); occult SUI exists only in the presence of reduced prolapse.

**Concept** — PFDs are common, under-reported and defined by **symptoms and bother, not anatomy**: half of parous women have anatomical prolapse, but only those with symptoms have the disease.

## DIAGNOSTIC EVALUATION

### General Considerations

- **Patient-reported and QoL outcomes** are now central; PFDs affect social, psychological, economic and sexual life, so **treatment success is hard to define**.
- **PFDs coexist and interact** — POP may disturb voiding or defecation.
- **Women delay seeking care** for 1 to 5 years or more after symptoms become bothersome and rarely volunteer them → **ask proactively**; a short validated screening questionnaire opens the discussion.
- **POP and UI are distinct diagnoses**: patients may expect prolapse repair to cure leakage. Relief of bulge drives satisfaction after POP surgery → **educate on anatomy and set expectations**.
- **Inequities** in care for minority women operate at patient, provider–patient and system levels; social determinants of health and systemic racism need evaluation.
- **Identify and quantify the UI type** to target treatment; consider reversible causes — **DIAPPERS** (Table 105.2).
- **POP compartments**: anterior (**cystocele** — bladder descent); posterior (**rectocele** — posterior wall protrudes); apical (**uterine prolapse**, vault, and **enterocele** — bowel at the vaginal apex).
- **Rectocele vs rectal prolapse**: rectocele is posterior vaginal wall into the vagina; rectal prolapse is **rectal mucosa out through the anus**.
- **Evaluation** → **differential** → **shared decision-making**; degree of bother balances risk against benefit, alongside the patient's goals.
- **Objective improvement does not always predict subjective improvement**. Good communication improves satisfaction, adherence, emotional health, symptom resolution and pain control.

**TABLE 105.2 CAUSES OF TRANSIENT URINARY INCONTINENCE (DIAPPERS)**

- Delirium · Infection · Atrophic vaginitis/urethritis · Psychological (severe depression, neurosis) · Pharmacological · Excessive urine production · Restricted mobility · Stool impaction.
- **What it is for**: a checklist of **reversible** causes to exclude before labelling UI as established — especially in the elderly. **Trap**: these are secondary causes, not UI subtypes.

From Resnick NM: Urinary incontinence in the elderly. Med Grand Rounds, 1984.

**FIG. 105.1 MANAGEMENT OF URINARY INCONTINENCE IN FRAIL OLDER INDIVIDUALS**

From Gibson W, et al.: Incontinence in frail elderly persons: report of the 6th International Consultation on Incontinence (ICI), 2021.

**Active case-finding in all frail elderly people** → **clinical assessment**: assess, treat and reassess treatable conditions (delirium, infection, pharmaceuticals, psychological, excess urine output, reduced mobility, stool impaction); comorbidities and activities of daily living (ADLs); QoL and the goals of patient and carer; targeted examination (cognition, mobility, neurological, digital rectal); urinalysis; bladder diary or wet checks (especially for nocturia); postvoid residual (PVR) in selected patients → **clinical diagnosis** → **initial management** (in mixed UI treat the most bothersome symptom first) → if insufficient improvement, reassess and treat comorbidity ± functional impairment → if still insufficient, or severe symptoms, **specialist referral** by preference and comorbidity.

| Clinical diagnosis                  | Initial management                                                                                                                                                                              |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Urgency UI                          | Lifestyle interventions; behavioural therapy; consider an antimuscarinic trial                                                                                                                  |
| Significant PVR                     | Treat constipation; review medications; alpha-blocker trial (men); <b>catheter drainage if PVR 200–500 mL</b> , then reassess                                                                   |
| Stress UI                           | Lifestyle interventions; behavioural therapy                                                                                                                                                    |
| Other (red flags) → direct referral | Pain; haematuria; recurrent symptomatic urinary tract infection (UTI); pelvic mass; pelvic irradiation; pelvic or lower urinary tract surgery; prolapse beyond the introitus; suspected fistula |

- Diagnoses overlap (mixed UI; **detrusor hyperactivity with impaired contractile function, DHIC**). **Do not treat asymptomatic bacteriuria.**

## History

- **History is the foundation**; women seeking help should be **believed** and their symptoms validated.
- **SUI is a urethral problem, UII a bladder problem, MUI both**; MUI may be stress- or urgency-predominant.
- **Accuracy of history**: systematic reviews show a positive SUI history predicts urodynamic SUI in about three-quarters, but a negative history does not exclude it → examination, voiding diary or urodynamic studies (UDS) if the type is unclear.

## History of Present Illness

- **Parameters**: type, duration, severity, frequency of episodes, volume, bother, pad use, trajectory (worse, better, stable); alleviating and aggravating factors; prior treatments and results; **treatment goals**.
- **Typing questions**: leakage with laughing, coughing or activity → SUI; leakage preceded by urgency or on the way to the toilet → UII; both → MUI.
- **Bother is assessed per type** — in MUI it decides which component is treated first.
- **Quantity**: pads per day, pad size, containment, clothing changes, volume (drops → tablespoons → whole bladder). **Pad weight testing** is used mainly in trials and academic settings.
- **Voiding habits**: hesitancy, incomplete emptying, stream strength, intermittency, straining, positional voiding, daytime and night-time frequency; **urgency and frequency** are the key lower urinary tract symptoms (LUTS) to characterise.
- **Onset**: traumatic delivery; lifelong since childhood; peri- or postmenopause; after surgery; neurological disease. **Urinary symptoms may precede a neurological diagnosis** → screen for numbness, tingling, tremor, balance or coordination loss, and blurred or double vision.
- **QoL impact**: patients modify lifestyle to avoid leakage; use questionnaires (see Questionnaires).
- **POP — bother is the key**. Stage 1 (within the vagina) may be asymptomatic; **symptoms usually begin when POP reaches or passes the hymen**, where it can be seen and felt: stage 2 (–1 to +1 cm), stage 3 (>1 cm beyond), stage 4 (complete eversion).
- **POP is dynamic** — worse later in the day and with activity.
- **POP symptoms**: bulge, pressure, heaviness, "falling out". **Splinting** to void or defecate suggests high-grade POP; high-grade anterior POP can **raise urethral resistance and impede voiding**.
- **POP is associated with DO and overactive bladder (OAB)**; repair may improve OAB, but not in every patient → **staged OAB treatment** may be needed.
- **SUI coexists in about 40% of women with POP**.

- **Posterior POP (rectocele):** obstructed defecation, bulge, splinting for bowel movements, stool trapping, dyspareunia, poor self-image, reduced vaginal sensation, a sense of a wide hiatus or weakness, a ball interfering with activity.

#### OCCULT VERSUS DE NOVO STRESS URINARY INCONTINENCE

|                    | Occult SUI                                                                                    | De novo SUI                                                                                       |
|--------------------|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| <b>Definition</b>  | <b>ICS-defined:</b> SUI appearing when POP is <b>reduced preoperatively</b>                   | <b>No ICS definition;</b> SUI developing <b>after POP surgery</b> with no preoperative occult SUI |
| <b>Frequency</b>   | About <b>one-third</b> of women undergoing vaginal POP repair                                 | Widely variable (from a few per cent to half of clinically continent women)                       |
| <b>Implication</b> | Counsel and share the decision on <b>concurrent versus staged</b> anti-incontinence treatment |                                                                                                   |

#### Compassionate Care

- **Compassion improves outcomes and lowers costs** — medical outcomes, parasympathetic responses, well-being, self-care, sense of purpose.
- A strong doctor–patient relationship **increases medication adherence**.
- **Compassion need not take time:** expressions lasting under 1 minute are meaningful — scripted reassurance by an oncologist reduced anxiety in women with breast cancer.
- Validate and acknowledge concerns → **patient-centred care**.

#### Past Medical and Surgical History

- **Urological:** infections, stones, haematuria, trauma, procedures (childhood and adult).
- **Neurological:** neurological trauma, stroke, spinal cord injury or surgery, back pain or surgery, multiple sclerosis, Parkinson disease, multiple system atrophy, Huntington disease, myelodysplasia.
- **Medical:** diabetes, thyroid disease, dementia; **prior pelvic radiation**.
- **Surgical:** pelvic trauma or surgery (including prior UI or POP surgery, hysterectomy) may damage sphincter or bladder function; **rectal resection** (low anterior, abdominoperineal) and **sacral lesion resection** can injure pelvic nerves → urinary, bowel, pelvic and sexual dysfunction.
- **Obstetric and gynaecological:** gravidity, number and mode of delivery, hormonal status (pre-, peri-, postmenopause; systemic or local hormone therapy; oral contraceptives; oophorectomy).
- **Obstetric risk factors for UI and POP:** prolonged second stage, large fetal head circumference, high birth weight, **forceps or vacuum delivery**.

|                           |                                                                   |
|---------------------------|-------------------------------------------------------------------|
| Local (vaginal) oestrogen | Systemic oestrogen (± progestogen, including oral contraceptives) |
| May improve UI            | May increase UI and urogenital atrophy symptoms                   |

#### Medications

- Drugs alter outlet resistance, bladder contractility, urine production or cognition; **stopping the culprit may reverse the dysfunction**.

TABLE 105.3 PHARMACOLOGIC AGENTS THAT MAY AFFECT LOWER URINARY TRACT FUNCTION

| Mechanism           | Class                 | Examples                                         | Effect                                                           |
|---------------------|-----------------------|--------------------------------------------------|------------------------------------------------------------------|
| ↑ Outlet resistance | Sympathomimetics      | Ephedrine, methylphenidate, cocaine, amphetamine | Obstructive symptoms and/or OAB                                  |
|                     | α-adrenergic agonists | Midodrine, pseudoephedrine                       | ↑ urethral (and prostatic capsule) tone → obstruction, retention |

| Mechanism                  | Class                                                | Examples                                                       | Effect                                               |
|----------------------------|------------------------------------------------------|----------------------------------------------------------------|------------------------------------------------------|
| ↓ <b>Outlet resistance</b> | Sympatholytics ( $\alpha$ -blockers)                 | Terazosin, doxazosin, tamsulosin, alfuzosin, silodosin         | <b>Exacerbate SUI</b>                                |
| ↓ <b>Emptying</b>          | Anticholinergics                                     | Oxybutynin, fesoterodine, solifenacin, trospium, darifenacin   | Retention, especially with outlet obstruction        |
|                            | Calcium-channel blockers                             | Amlodipine, diltiazem                                          | Impaired emptying, retention, constipation           |
|                            | Opioids                                              | Oxycodone, morphine                                            | Retention, constipation, confusion, immobility       |
|                            | Gabapentin                                           | —                                                              | Impaired bladder emptying                            |
| ↑ <b>Contractility</b>     | Cholinesterase inhibitors                            | Donepezil, galantamine, rivastigmine                           | <b>Precipitate UUI</b>                               |
|                            | Selective serotonin reuptake inhibitors (SSRIs)      | Fluoxetine, sertraline                                         | ↑ cholinergic transmission → UI                      |
| ↑ <b>Urine output</b>      | Diuretics                                            | Furosemide, thiazides, spironolactone, triamterene, bumetanide | Frequency, UI                                        |
|                            | Lithium                                              | —                                                              | Polyuria (nephrogenic diabetes insipidus)            |
|                            | Sodium–glucose cotransporter 2 (SGLT2) inhibitors    | Canagliflozin, dapagliflozin, empagliflozin                    | Glycosuria, polyuria, ↑ UTI                          |
|                            | Glitazones                                           | Pioglitazone, rosiglitazone                                    | Oedema → nocturnal polyuria, nocturia, night-time UI |
| <b>Indirect</b>            | Psychotropics (sedatives, hypnotics, antipsychotics) | Diazepam, olanzapine                                           | Confusion, impaired mobility                         |
|                            | Angiotensin-converting enzyme (ACE) inhibitors       | Lisinopril, enalapril                                          | <b>Cough</b> → worsens SUI                           |

From Gibson W, et al.: 6th International Consultation on Incontinence, 2021.

- **Pattern:** every agent acts through one of four routes — outlet, detrusor, urine volume, or the patient's cognition and mobility.
- **Traps:**  $\alpha$ -blockers (usually thought of as helping voiding) **worsen SUI** in women; ACE inhibitors cause UI via **cough**; dementia drugs (cholinesterase inhibitors) and antimuscarinics act in opposite directions on the detrusor.

### Other Contributing Factors

- **Family history of POP** → higher risk of POP and of recurrence; connective-tissue genetics (collagen, laminins, elastic fibres) implicated. Genetics also contribute to OAB, SUI and UUI.
- **Social history:** smoking, substance use, sexual activity, partner, social support, physical activity, occupation.
- **Screen for pelvic trauma and abuse** — trauma is common; asking creates a safe, healing environment (trauma-informed care).

## Social Determinants of Health

- **Social determinants of health (SDOH)** = conditions where people live and work that affect health and QoL; assess them in history-taking. They drive health disparities, which are a barrier to equity and an economic burden.

### KEY POINTS

- PFDs are common and rising; history and examination are central; bother guides therapy.
- Adjuncts: UA, PVR, UDS, cystoscopy, imaging, questionnaires. Surgical and non-surgical options exist.
- POP becomes symptomatic **at or beyond the vaginal opening**; screen for pelvic and sexual trauma.

**Concept** — the history sorts leakage into **urethral (stress) versus bladder (urgency) failure** and prolapse into anatomy versus bother; it is the **patient's bother, not the finding**, that decides what is treated and in what order.

## PHYSICAL EXAMINATION

- **General:** age, mobility and frailty inform performance status. **Abdomen:** habitus, incisions, hernias, organomegaly, bladder distension — relevant if abdominal surgery is considered.
- **Frailty predicts poor postoperative outcomes and complications** in urological surgery → **preoperative frailty assessment recommended at age ≥65**.
- **Frailty is distinct from comorbidity:** lack of resilience to external stressors, or decline in physiological reserve. **No single standard definition or tool.**

### FRAILITY ASSESSMENT TOOLS

| Tool                         | What it measures                                                                                              |
|------------------------------|---------------------------------------------------------------------------------------------------------------|
| Fried Frailty Index          | Five domains: shrinking (weight loss), physical energy, activity, <b>grip strength</b> , <b>walking speed</b> |
| Timed Up and Go Test         | Timed functional mobility                                                                                     |
| Clinical Frailty Scale (CFS) | Nine-point visual pictorial scale — <b>rapid</b> bedside assessment                                           |

- **External genitalia:** appearance, lesions, labial atrophy or adhesions, **oestrogen status** — judged by urethral caruncle, urethral prolapse, labial adhesions, colour and rugae.

| Hypo-oestrogenic (atrophic) tissue             | Well-oestrogenised tissue |
|------------------------------------------------|---------------------------|
| Pale, flat, dry, no rugae; may be erythematous | Pink and rugated          |

- The 2021 Centers for Medicare & Medicaid Services (CMS) evaluation and management guidelines no longer require bulleted examination elements — but the **pelvic examination remains irreplaceable**.

TABLE 105.4 COMPONENTS OF THE FEMALE GENITOURINARY EXAMINATION

| Structure          | Assess                                                                          |
|--------------------|---------------------------------------------------------------------------------|
| External genitalia | Appearance, hair distribution, lesions                                          |
| Urethral meatus    | Size, location, lesions, prolapse                                               |
| Urethra            | Masses, tenderness, scarring                                                    |
| Bladder            | Fullness, masses, tenderness                                                    |
| Vagina             | Appearance, oestrogen effect, discharge, lesions, support, cystocele, rectocele |
| Cervix             | Appearance, lesions, discharge                                                  |

| Structure             | Assess                                                              |
|-----------------------|---------------------------------------------------------------------|
| Uterus                | Size, contour, position, mobility, tenderness, consistency, descent |
| Adnexa and parametria | Masses, tenderness, organomegaly, nodularity                        |
| Anus and perineum     | —                                                                   |

From 1997 Documentation Guidelines for Evaluation and Management Services (American Medical Association and HCFA), revised November 1997.

- **Use:** a systematic, outside-in sequence so no compartment is missed.

### Physical Examination for Stress Urinary Incontinence

- **2023 American Urological Association (AUA) / Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction (SUFU) guideline:** objectively demonstrate SUI with a comfortably full bladder before surgery. Supine or standing; **stand the patient if supine testing is negative.**
- **Positive cough stress test (CST) = objective confirmation of SUI.**

| CST at half functional capacity                                                                                                           | CST at comfortably full bladder                                            |
|-------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Functional capacity = largest voided volume on the diary; sitting or standing. <b>High specificity and sensitivity</b> for urodynamic SUI | Sitting or standing. Similar sensitivity but <b>much lower specificity</b> |

- **With POP, test with the anterior wall supported** — a large prolapse **kinks the urethra and masks SUI.**
- **Assess urethral position and mobility** at rest, on Valsalva and on coughing.

**Q-tip test:** lubricated cotton swab into the urethra to the urethrovesical junction → Valsalva → measure angle moved from horizontal → **hypermobility = >30°.**

#### TESTS OF URETHRAL MOBILITY

| Test                                        | Method                                                           | Place                                                      |
|---------------------------------------------|------------------------------------------------------------------|------------------------------------------------------------|
| Q-tip test                                  | Intraurethral swab; angle measured                               | Historical; uncomfortable; <b>trials and teaching only</b> |
| Vaginal swab test                           | Swab in the vagina                                               | <b>Equivalent</b> to Q-tip, less discomfort                |
| Visual urethral mobility examination (VUME) | Meatus observed on cough and Valsalva; rotation judged < or >30° | <b>Routine clinical practice</b> — feasible, accurate      |

### Physical Examination for Pelvic Organ Prolapse

- **Examine in lithotomy and standing** (one foot on a step) to reproduce the **greatest degree** of prolapse the patient experiences.
- Assess each compartment — **anterior, posterior, apical** (cervix or cuff) — and the **perineal body**, for position relative to the introitus and for laxity.

**Split Graves speculum technique:** one posterior blade retracts the posterior wall → examine anterior compartment → reposition blade anteriorly → examine posterior compartment → insert both blades (anterior and posterior) → isolate apex (cuff or cervix) → withdraw posterior blade slowly → examine posterior wall → with posterior blade in place, patient strains.

| Blade expelled on straining                                                                                         | Blade stays in place                                           |
|---------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| Posterior wall foreshortens → <b>loss of level I support</b> (cardinal–uterosacral complex) → <b>vault prolapse</b> | <b>Isolated rectocele or enterocele</b> without vault prolapse |

- **Levels of support I, II and III** are those described by **DeLancey** (1992); level I = cardinal–uterosacral apical support.

- **Most widely used classifications: Baden-Walker and POP-Q** (Fig. 105.2).

FIG. 105.2 COMPARISON OF GRADING SYSTEMS USED TO QUANTIFY PELVIC ORGAN PROLAPSE

| Level of descent (on straining) | Porges (1963) severity | Baden (1972) vaginal profile | Beecham (1980) | POP-Q (1996; ICS, AUGS, SGS)                     |
|---------------------------------|------------------------|------------------------------|----------------|--------------------------------------------------|
| Upper vagina → midplane         | Slight / 1st degree    | Grade 1                      | 1st degree     | Stage I                                          |
| Midplane → hymen                | Slight / 1st degree    | Grade 2                      | 1st degree     | Stage I, then <b>stage II from -1 cm</b>         |
| Through and beyond hymen        | Moderate / 2nd degree  | Grade 3                      | 2nd degree     | Stage II to +1 cm; <b>stage III beyond +1 cm</b> |
| <b>Complete eversion</b>        | Marked / 3rd degree    | Grade 4                      | 3rd degree     | Stage IV                                         |

From Theofrastous JP, Swift SE: The clinical evaluation of pelvic floor dysfunction. Obstet Gynecol Clin North Am, 1998. AUGS, American Urogynecologic Society; SGS, Society of Gynecologic Surgeons.

- **What the figure tests:** older systems are **ordinal and hymen-/introitus-referenced by eye**; POP-Q is **quantitative, in centimetres**, and alone gives a **2-cm window (-1 to +1)** around the hymen as stage II.
- **POP-Q** was created for objectivity: **nine points** measured relative to the **hymenal ring**.
- **Six vaginal points** (Aa, Ba, C, D, Ap, Bp) measured **on Valsalva**; **above the hymen negative, beyond positive, at the hymen zero**.
- **Three lengths:** genital hiatus (gh), perineal body (pb), and total vaginal length (tvL, measured with the prolapse **reduced**).

FIG. 105.3 POP-Q POINTS OF REFERENCE

| Point      | Definition                                                    | Range         |
|------------|---------------------------------------------------------------|---------------|
| <b>Aa</b>  | Anterior wall <b>3 cm proximal to the hymen</b>               | -3 to +3 cm   |
| <b>Ba</b>  | Most distal point of the remaining upper anterior wall        | -3 cm to +tvL |
| <b>C</b>   | Most distal edge of cervix or vaginal cuff scar               | —             |
| <b>D</b>   | Posterior fornix ( <b>not applicable after hysterectomy</b> ) | —             |
| <b>Ap</b>  | Posterior wall 3 cm proximal to the hymen                     | -3 to +3 cm   |
| <b>Bp</b>  | Most distal point of the remaining upper posterior wall       | -3 cm to +tvL |
| <b>gh</b>  | Middle of external urethral meatus → posterior midline hymen  | —             |
| <b>pb</b>  | Posterior margin of gh → middle of anal opening               | —             |
| <b>tvL</b> | Vaginal depth with C or D reduced to normal position          | —             |

From Bump RC, et al.: The standardization of terminology of female pelvic organ prolapse and pelvic floor dysfunction. Am J Obstet Gynecol, 1996.

- **Logic:** A points are **fixed** (3 cm in), so they range only  $\pm 3$  cm; B points are **mobile** (the leading part of each wall), so they can reach +tvL. With no prolapse, Aa, Ba, Ap and Bp all sit at -3.

TABLE 105.5 POP-Q STAGING CRITERIA

| Stage      | Criteria                                                       |
|------------|----------------------------------------------------------------|
| <b>0</b>   | Aa, Ap, Ba, Bp at -3 cm <b>and</b> C or D $\leq$ -(tvL - 2) cm |
| <b>I</b>   | Stage 0 criteria not met; leading edge < -1 cm                 |
| <b>II</b>  | Leading edge $\geq$ -1 cm but $\leq$ +1 cm                     |
| <b>III</b> | Leading edge > +1 cm but < +(tvL - 2) cm                       |

| Stage | Criteria                                                     |
|-------|--------------------------------------------------------------|
| IV    | Leading edge $\geq +(\text{tvI} - 2)$ cm (complete eversion) |

From Bump RC, et al., 1996.

- **Stage is set by the single most distal point** (leading edge) in any compartment.
- **Trap:** the stage III/IV boundary is **relative to tvI**, not a fixed number; stage 0 allows 2 cm of apical slack (C or D within  $\text{tvI} - 2$ ).

FIG. 105.4 WORKED EXAMPLE — POSTERIOR SUPPORT DEFECT

| Aa | Ba | C  | gh  | pb | tvI | Ap | Bp |
|----|----|----|-----|----|-----|----|----|
| -3 | -3 | -6 | 4.5 | 1  | 8   | +2 | +5 |

- **Anterior compartment well supported** (Aa, Ba at -3). **Bp is the leading edge, 5 cm beyond the hymen.** With tvI 8 cm, C at -6 means the **cuff has descended 2 cm**. Leading edge +5 lies between +1 and  $+(8 - 2) = +6 \rightarrow$  **stage III**.

*Prior mesh.*

- **Prior polypropylene mesh or implant with pain or voiding symptoms**  $\rightarrow$  thorough history (device type, timing of symptom onset) and examination; some patients are asymptomatic.
- **Examine for** presence, location and size of extrusion or exposure; **document tenderness even without visible exposure.**
- **Adjuncts:** cystoscopy, digital rectal examination (DRE), translabial ultrasound (US; mesh relative to urethra and bladder), magnetic resonance imaging (MRI).
- Classify with the **2011 International Urogynecological Association (IUGA)/ICS joint terminology** (Table 105.6).

TABLE 105.6 INTERNATIONAL UROGYNECOLOGICAL ASSOCIATION/INTERNATIONAL CONTINENCE SOCIETY JOINT TERMINOLOGY AND CLASSIFICATION OF THE COMPLICATIONS RELATED DIRECTLY TO THE INSERTION OF PROSTHESES (MESHES, IMPLANTS, TAPES) AND GRAFTS IN FEMALE PELVIC FLOOR SURGERY

| Term                         | Definition                                                                                                        |
|------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Complication</b>          | Morbid process or event during surgery that is not an essential part of it                                        |
| <b>Contraction</b>           | Shrinkage or reduction in size                                                                                    |
| <b>Prominence</b>            | Protrusion beyond the surface (wrinkling, folding) with <b>no epithelial separation</b>                           |
| <b>Separation</b>            | Physically <b>disconnected</b> (e.g., vaginal epithelium)                                                         |
| <b>Extrusion</b>             | Passage <b>gradually out of</b> a body structure or tissue                                                        |
| <b>Compromise</b>            | Bring into danger                                                                                                 |
| <b>Perforation</b>           | Abnormal opening <b>into a hollow organ</b> or viscus                                                             |
| <b>Dehiscence</b>            | Bursting open or gaping along a natural or sutured line                                                           |
| <b>Sinus tract formation</b> | (Localised) fistulous tract towards vagina or skin, <b>with no visible implant</b> in the lumen or overlying skin |

From Haylen BT, et al.: IUGA/ICS joint terminology... Int Urogynecol J, 2011.

- **Contrasts tested:** **prominence** (no epithelial breach) vs **separation** (epithelial breach); **extrusion** (out through vaginal epithelium) vs **perforation** (into a viscus such as bladder or urethra); **sinus tract** = fistulous track *without* visible mesh.

*Neurological examination.*

- **Perform in any patient with known or suspected neurological disease:** speech, gait, cognition, facial symmetry, lower-limb and perineal/perianal sensation, lower-limb power, vaginal and pelvic floor strength. **Mobility and cognition** govern timely toilet access.
- **Bulbocavernosus reflex (BCR) tests S2–S4; present in 70% of normal women vs 100% of normal men** → an absent BCR in a woman is not by itself pathological.

**Eliciting the BCR:** squeeze the clitoris (or glans) → anal and pelvic floor contraction, seen or felt on DRE; alternatively, pull an indwelling catheter balloon against the bladder neck.

- **Anal sphincter tone (also S2–S4):** ask the patient to tighten as if stopping the stream. Laxity = neurological deficit **or** poor voluntary muscle awareness.
- **Urethra:** meatal stenosis, mass, tenderness, fullness, foreign body; expect postsurgical changes after anti-incontinence surgery or a mesh mid-urethral sling (MUS).

**Trauma-Informed Care**

- **Trauma-informed care** = awareness of the patient's whole life situation; improves engagement and adherence. Recognise trauma in patients, families and staff; **avoid retraumatisation**; embed in policy.
- **Sexual abuse is common:** about **1 in 4 women and 1 in 13 men** in the USA experience child sexual abuse; rape-related post-traumatic stress disorder (PTSD) affects millions of American women.
- **Genital examination triggers flashbacks or anxiety in about half of trauma survivors.**
- **Principles — safety, control and choice:** patient chooses timing (now or later) and a support person; may self-insert or decline the speculum; clinician **leaves the room while the patient undresses**; privacy prioritised; expose only what is needed for the shortest time; **discuss while the patient is dressed**.

**Chaperones During Physical Examinations**

- **Role:** comfort, safety, privacy and dignity for patient and provider; positioned to observe and assist; prevents misunderstandings. Supported by gowns, private undressing, sensitive draping, explanation.
- **American Medical Association (AMA) guidance** — physicians should:
  - Adopt and communicate a policy that **patients may request a chaperone; always honour** the request.
  - Use an **authorised member of the health-care team**, bound by privacy and confidentiality.
  - **Use a chaperone even when a trusted companion is present.**
  - Offer private conversation without the chaperone; minimise sensitive history-taking during the chaperoned examination.
- Patients and physicians differ on the role and choice of chaperone; **offering one is respectful** of privacy and autonomy.

**KEY POINTS**

- Systematic genitourinary examination; POP supine *and* standing; SUI supine or standing.
- Baden-Walker and POP-Q most used; neurological and rectal examination when indicated.
- Trauma-informed principles; offer a chaperone.

**Concept** — the examination is designed to **reproduce the patient's worst state** — full bladder, standing, straining, prolapse reduced — because each omission (empty bladder, supine position, unreduced kinking prolapse) **hides the pathology** the patient actually lives with.

## SUPPLEMENTAL EVALUATION

- **Urinalysis (UA) and PVR should be performed in most patients evaluated for UI.** No agreed standard exists for cystoscopy, UDS, imaging, diaries, pad tests or questionnaires → AUA/SUFU guidelines.
- **Weigh the value and accuracy** (sensitivity, specificity) of any extra test against its **cost and morbidity**.

### GUIDELINE RECOMMENDATIONS FOR INITIAL EVALUATION

|                               | AUA guideline — non-neurogenic OAB                            | AUA/SUFU 2023 — index patient wanting SUI surgery                              |
|-------------------------------|---------------------------------------------------------------|--------------------------------------------------------------------------------|
| <b>Required</b>               | <b>UA</b>                                                     | <b>UA and PVR</b> ; objective demonstration of SUI                             |
| <b>Clinician's discretion</b> | Urine culture, PVR, bladder diary, symptom questionnaires     | —                                                                              |
| <b>Not in initial work-up</b> | UDS, cystoscopy, renal and bladder US (uncomplicated patient) | Cystoscopy, UDS, imaging — <b>only in specific circumstances</b> (Table 105.7) |

- **Also consider further testing** with concomitant OAB, failed anti-incontinence surgery, or prior POP surgery.
- **Trap:** PVR is **required** for the SUI surgical candidate but only **optional** in OAB.

TABLE 105.7 CIRCUMSTANCES THAT WARRANT CONSIDERATION OF SUPPLEMENTAL EVALUATION IN THE SETTING OF STRESS URINARY INCONTINENCE

- **Diagnostic uncertainty:** no definitive diagnosis from symptoms and initial evaluation; **SUI not demonstrable**.
- **Bladder or neurological confounders:** known or suspected **neurogenic** lower urinary tract dysfunction; urgency-predominant MUI.
- **Abnormal basic tests:** abnormal UA (unexplained haematuria or pyuria); **elevated PVR** (clinician judgement); significant voiding dysfunction.
- **High-grade POP (POP-Q stage ≥3)** when SUI is not demonstrated on prolapse reduction.
- **What the table is for:** the "index patient" is uncomplicated; **any deviation from index status** — uncertainty, neurology, urgency, retention, prolapse — earns further testing.

## Symptom Quantification Instruments

### Bladder Diary

- **Bladder diaries** (voiding or micturition diaries, frequency-volume charts) quantify UI and are **both diagnostic and therapeutic** — patients gain insight into behaviours driving symptoms. Full discussion lies in the online chapter and Chapter 114.

### Questionnaires

- **Severity alone does not show QoL impact;** POP and UI are not life-threatening but can severely impair QoL → assess self-reported symptoms and their impact.
- **International Consultation on Incontinence Questionnaire (ICIQ):** modular, validated, international standard for clinical and research use, in over 30 languages. Modules: UI short form (UI SF), bladder diary, male/female LUTS (MLUTS/FLUTS) and their sexual modules, nocturia (N), OAB, vaginal symptoms (VS); **in development:** cognitively impaired adults (Cog), underactive bladder (UAB), children (CLUTS).
- **ICIQ-UI SF correlates with 1-hour and 24-hour pad tests** for SUI severity, and its change correlates with the Patient Global Impression of Improvement (PGI-I) after sling surgery.
- **Validated questionnaires are encouraged** to standardise patient-reported outcomes; choose by clinical question. **No single universal instrument**, making comparison hard — but validated tools correlate well.

**TABLE 105.8 INSTRUMENTS HIGHLY RECOMMENDED BY THE INTERNATIONAL CONSULTATION ON INCONTINENCE GROUP FOR URINARY INCONTINENCE, OVERACTIVE BLADDER, LOWER URINARY TRACT SYMPTOMS, AND PELVIC ORGAN PROLAPSE PATIENT-REPORTED OUTCOMES**

| Domain                            | Instruments (population)                                                                                                                                                            | Distinguishing purpose                                                                                                                                       |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Health-related QoL (HRQoL)</b> | BFLUTS/ICIQ-FLUTS (F); <b>ICIQ-UI SF</b> (M+F); IIQ and IIQ-7 (F); I-QoL (F, UUI); <b>KHQ</b> (M+F, OAB); N-QoL (nocturia); OAB-q and OABq-SF (M+F); PRAFAB (F, UUI); UISS (F, UUI) | IIQ — primarily SUI impact; OAB-q — OAB wet and dry; PRAFAB (Protection, Amount, Frequency, Adjustment, Body image) — UUI treatment effect; N-QoL — nocturia |
| <b>Screeners</b>                  | B-SAQ (F); LUSQ (storage LUTS); OAB-SS (7 items); <b>OAB-V8</b> (8 items); <b>QUID</b> (6 items, F); ICIQ bladder diary                                                             | OAB-V8 — <b>primary care</b> OAB screen; QUID — <b>diagnoses SUI vs UUI</b> ; ICIQ diary — <b>3-day</b> volume, frequency, intake, leaks                     |
| <b>Symptom bother</b>             | PPBC (M+F); UDI-6 (F)                                                                                                                                                               | PPBC — global impression of bladder condition                                                                                                                |
| <b>Urgency</b>                    | IUSS (Indevus Urgency Severity Scale)                                                                                                                                               | Urgency grade per void, in diaries                                                                                                                           |
| <b>POP symptoms and QoL</b>       | <b>PFDI</b> (Pelvic Floor Distress Inventory); <b>PFIQ</b> (Pelvic Floor Impact Questionnaire)                                                                                      | PFDI — <b>symptoms</b> ; PFIQ — <b>QoL impact</b>                                                                                                            |
| <b>Sexual matters</b>             | ICIQ-FLUTSsex; ICIQ-Bsex (F)                                                                                                                                                        | Sexual bother from LUTS and bowel symptoms                                                                                                                   |

From Abrams P, et al.: 6th International Consultation on Incontinence, Neurourol Urodyn, 2018. F, female; M, male.

- **Pattern:** *distress/symptom* instruments (UDI, PFDI) pair with *impact* instruments (IIQ, PFIQ). **Trap:** PFDI measures prolapse **symptoms**, PFIQ its **impact**; UDI-6 is bother, IIQ-7 is impact — both short forms.

### Nomograms and Risk Calculators

- Risk models exist, though their use in practice is uncertain: **postpartum UI** (primiparous and multiparous); postpartum pelvic floor US predicting SUI; **de novo SUI after POP surgery**; long-term risk of POP, UI and FI (UR-CHOICE). Cleveland Clinic calculators (riskcalc.org) aid **counselling**.

### Pad Tests

- **Non-invasive objective quantification of UI**, used chiefly in trials. Protocols: 20-minute, standardised 1-hour, 24-hour, 48-hour — **shorter tests use set activities but give a limited picture; longer tests are more reproducible but less complied with.**
- **ICS: 3-day bladder diary and pad weight test** may both be used in incontinence research. 24-hour testing suffices clinically; longer for research; 1-hour tests typically at initial evaluation.

**ICS standardised 1-hour pad test:** preweighed pad, no voiding → drink **500 mL sodium-free fluid in <15 min**, then rest → walk **30 min** including one flight of stairs up and down → stand from sitting ×10, cough vigorously ×10, run on the spot 1 min, bend to pick up an object ×5, wash hands in running water 1 min (adjusted to fitness) → weigh pad. **If the patient must void, restart the test.**

|                                          | 1-hour pad test                 | 24-hour pad test                                                                                |
|------------------------------------------|---------------------------------|-------------------------------------------------------------------------------------------------|
| <b>Conduct</b>                           | Standardised activities (above) | Start with <b>empty bladder</b> ; normal activities; weigh immediately or store in airtight bag |
| <b>Non-urine gain</b> (sweat, discharge) | <1.4 g                          | <4.4 g                                                                                          |
| <b>Mild</b>                              | 1–10 g                          | 4–20 g                                                                                          |
| <b>Moderate</b>                          | 11–50 g                         | 21–74 g                                                                                         |
| <b>Severe</b>                            | >50 g                           | ≥75 g                                                                                           |

|                           | 1-hour pad test                                          | 24-hour pad test                                                                                                         |
|---------------------------|----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>Strengths / limits</b> | Bladder volume not standardised (no consensus on volume) | <b>More reproducible, more clinically useful</b> ; affected by compliance, intake, activity, hormones, heat and humidity |

- **Inconclusive small gains** → oral **phenazopyridine** stains urine orange to confirm it is urine.
- **24-hour pad test does not predict urodynamic SUI**, but correlates with incontinence episode frequency on a 3-day diary.
- **Trap: pad counts** (recalled or collected) **do not correlate** with severity on 24-hour pad weight.

## Dye Testing

| Test                                                                               | Stain                                                      | Interpretation                                              |
|------------------------------------------------------------------------------------|------------------------------------------------------------|-------------------------------------------------------------|
| <b>Oral phenazopyridine</b> 100–200 mg three times daily                           | Urine orange                                               | Confirms leaking fluid is <b>urine</b>                      |
| <b>Intravesical methylene blue</b> (or phenazopyridine in sterile water or saline) | Tampon blue (or orange)                                    | <b>Vesicovaginal or urethrovaginal fistula</b>              |
| <b>Intravesical methylene blue + oral phenazopyridine</b> (double-dye tampon test) | <b>Orange</b> → ureter; <b>blue</b> → bladder; both → both | <b>Ureterovaginal</b> vs vesicovaginal fistula, or combined |

- **Logic:** oral dye reaches the vagina only via the kidney–ureter route; bladder dye only via the bladder — so the colour **localises the fistula**.

## Urinalysis

- **UA is part of the diagnostic evaluation of LUTS and UI:** haematuria, pyuria, glucosuria or proteinuria may indicate a **secondary cause**. Abnormal dipstick → microscopy or culture → further testing or treatment.

## Postvoid Residual

- **PVR** = urine remaining shortly after voiding; measures emptying. **US preferred over catheterisation**; portable bladder scanners are reliable, though devices vary.
- **Recommended by the European Association of Urology (EAU), ICS and AUA/SUFU** — yet evidence linking PVR to dysfunction is weak:
  - Slow stream, straining and incomplete-emptying symptoms **do not correlate** with measured PVR.
  - Voiding dysfunction on UDS or uroflow may coexist with **normal PVR**; patients with and without LUTS have similar PVRs.
- **A single PVR is unreliable — repeat it.** A single low PVR does not exclude incomplete emptying; a single high PVR does not predict emptying dysfunction.
- **No agreed cut-off** — thresholds range **30–200 mL**. In asymptomatic women almost all PVRs are **<100 mL** and few exceed 50 mL; defining normal in asymptomatic women may set cut-offs.
- **Elevated PVR** does not identify the cause but **warrants further evaluation** and alters counselling before **SUI procedures** or **drugs that reduce detrusor contractility** for UI; it may contribute to UTI, renal dysfunction and hydronephrosis.

## Cystoscopy

- **Not routine in the index SUI patient**; consider when an anatomical abnormality is suspected — urgency, haematuria, irritative LUTS.

- **After anti-incontinence surgery, mesh MUS or mesh reconstruction:** look for **foreign body** (mesh, suture) in urethra or bladder. **After pelvic radiation:** radiation change.
- **Tumours, stones, cystitis and foreign bodies** cause irritative LUTS, recurrent UTI and incontinence.
- **Survey:** ureteric orifices (morphology, position, number, efflux); mucosa — **trabeculation** (outlet obstruction and/or DO) and oestrogen status; urethra — foreign body, stricture, diverticulum, fistula, position.
- **Yield in low-risk preoperative women is low** (occasional early urothelial carcinoma, calculi, absent ureteric efflux) → **cystoscopy not required** before UI or POP surgery without another indication.

#### KEY POINTS

- History alone may mislead; UA and PVR in most; diaries (diagnostic and therapeutic), pad tests and validated questionnaires quantify symptoms and QoL.

*Concept — only UA and PVR are universal; every other test is **earned by a departure from the index patient**, and each must repay its cost and morbidity with information that changes management.*

## URODYNAMIC STUDIES

- Full discussion of UDS lies in the online chapter and Chapter 107.

### Radiographic Imaging

#### Voiding Cystourethrogram

- **Imaging is not needed in uncomplicated UI;** image if **renal damage or pelvic pathology** is suspected.
- **Voiding cystourethrogram (VCUG):** suspected **vesicoureteric reflux (VUR)** or **urethral diverticulum**; shows bladder contour and diverticula, bladder-neck configuration on voiding, bladder-base position on straining (cystocele grade), and fluoroscopically **subtle leakage on cough or Valsalva**.

#### Ultrasonography (Including Translabial Ultrasound to Visualise Mesh)

- **Renal US** — non-invasive assessment of obstruction and hydronephrosis. **Upper-tract imaging indications:** chronic retention, **severe POP** (ureteric kinking → hydroureteronephrosis), suspected fistula, neurogenic detrusor dysfunction.
- **Bladder and urethral US:** bladder wall thickness associates with OAB; in SUI, bladder-neck and urethral swing, bladder-neck funnelling, urethral position relative to the symphysis and hiatal area differ — **role in management still unclear**.
- **Translabial US for POP** identifies involved structures (rectocele vs enterocele); three-dimensional translabial US detects **levator ani defects** with moderate-to-good agreement with MRI — defects associate with POP and recurrence. **Translabial US ≠ pelvic US** (uterus and adnexa).
- **Translabial US for mesh:** the **only modality that reliably images mesh** and confirms the presence and location of a mesh MUS — proximity to urethra and bladder, proximal or distal position, path, contour, twisting, continuity after revision → plans revision or removal; helpful when mesh is impalpable or not seen cystoscopically. Office or theatre; distinct protocol.
- **Computed tomography (CT) and MRI are poor for mesh** — seen only if infected, associated with other pathology, or inferred from surrounding structures.

### Magnetic Resonance Imaging

- **MRI** gives soft-tissue detail of bladder neck, urethra, pelvic floor relaxation and prolapse.

- **MRI is the preferred, standard-of-care modality for urethral diverticulum.** (Full discussion online and Chapter 122.)
- **Classic triad — the three Ds: dysuria, postvoid dribbling, dyspareunia** — but often absent. Non-specific presentations: frequency, urgency, recurrent UTI, haematuria, tender mass, stones, purulent discharge, retention.
- **HMO system** (H-line, M-line, organ prolapse) on a midsagittal image at **maximal strain**:
  - **H-line** = levator hiatus width: inferior pubic symphysis → posterior levator plate.
  - **M-line** = muscular floor relaxation: descent of the levator plate from the **pubococcygeal line** (inferior symphysis → first/second coccygeal junction).
  - **O** = organ prolapse beyond the H-line; **POP on MRI = any organ beyond the H-line**, in centimetres.
- **Dynamic MRI indications:** complex patients, **symptoms out of proportion to examination**, prior repairs; with a POP protocol, confirms enterocele and rectocele and plans surgery.

#### IMAGING MODALITIES IN PELVIC FLOOR EVALUATION

| Modality          | Main indication                                                                                 |
|-------------------|-------------------------------------------------------------------------------------------------|
| VCUG              | VUR, urethral diverticulum, bladder diverticula, cystocele, subtle SUI                          |
| Renal US          | Hydronephrosis — retention, severe POP, fistula, neurogenic detrusor                            |
| Translabial US    | <b>Mesh (modality of choice)</b> ; compartment and levator defects                              |
| CT                | Poor for mesh                                                                                   |
| MRI / dynamic MRI | <b>Urethral diverticulum (standard of care)</b> ; complex or recurrent POP, discordant symptoms |

#### FIG. 105.5 MRI OF A PATIENT WITHOUT PROLAPSE

- Demonstrates the H-line (drawn along the puborectal line, PRL), the M-line, and the pubococcygeal line (PCL); no organ crosses the H-line. From Comiter CV, et al.: Urology, 1999.

**Concept** — each modality is chosen for the **one structure it sees best**: translabial US for synthetic mesh, MRI for soft tissue (diverticulum, dynamic prolapse), renal US for the upper tract — cross-sectional CT adds little.

## MANAGEMENT

### Urinary Incontinence Treatment Overview

- **Complete evaluation** = history and examination, plus exclusion of UTI, outlet obstruction, stones, foreign body or tumour where suspected.
- **UI arises from the bladder, the outlet or both**; treatment targets the cause, individualised to **patient goals, risk-benefit and cost-benefit**, with counselling to align them.
- Full treatment discussion lies in the online chapter and Chapters 113–118.

## CONCLUSIONS

- **Goal of reconstruction:** restore normal anatomy and function of vagina, bladder and surrounding structures to **improve QoL while minimising complications**.
- Pelvic floor care is refocusing on **patient experience, goals and QoL** and patient-centred outcomes research; the **patient-surgeon relationship** — open communication, trust, mutual respect — is pivotal.